



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

04-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	C	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C	A,B,C	A,B,D	A,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	8	3	2	5	5	7	1

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C,D	A,B,C,D	A,B,C	A,B,D	A,B	B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	8	8	3	2	3	5	4	5

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	B	B	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B,C	A,C,D	C,D	A,B,D	B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	7	4	5	5	1	1	1	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (D)**

I is the intensity of incident beam ab. The interfering waves are bc and ef, reflected from surface of I and II plate, respectively.

Reflection coefficient of intensity,

$$r = 25\% = 0.25$$

Transmission coefficient of intensity,

$$t = 75\% = 0.75$$

$$\text{The intensity of beam bc, } I_1 = 0.25 \quad I = \frac{1}{4}I$$

$$\text{The intensity of beam bd} = 0.75 I$$

$$\text{The intensity of beam de} = 0.25 \times 0.75I$$

The intensity of beam ef,

$$I_2 = 0.75 \times 0.25 \times 0.75I = \frac{9}{64}I$$

Ratio of maximum and minimum intensities,

$$\sqrt{\frac{I_{\max}}{I_{\min}}} = \frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} = 7$$

2. **Ans (C)**

The submerged sphere is subject to a buoyant force equal to the weight of the water in the volume displaced by the sphere. Let us call this weight of water w . One might be tempted to say that to restore equilibrium, a weight w should be added to the pan with the stand. However, according to Newton's third law, the force with which the water in the container acts on the submerged sphere is exactly equal to the water in the opposite direction. Hence, as the weight of the pan with the stand decreases, the weight of the pan with the container increases. Therefore, to restore balance, a weight equal to $2w$ must be placed on the pan with the stand. By the way, the tip of the balance does not indicate unequal torques. Two objects can balance at any angle of tip.

3. **Ans (B)**

Let the speed of wave be v , for crossing one wave crests to the other while travelling in the same direction, the surfing speed has to be greater than speed of the wave, i.e., $v < 15$ m/s.

Let wavelength of wave be λ m

While surfing in the same direction,

$$\lambda = (15 + v) \times 0.8$$

While surfing in the direction opposite to the wave motion,

$$\lambda = (15 - v) \times 0.6$$

$$(15 - v) 0.8 = (15 + v) 0.6$$

$$v = 15/7 \text{ m/s} = 2.143 \text{ m/s}$$

$$\text{So, } \lambda = (15 - 2.143) \times 0.8 = 10.3 \text{ m}$$

4. **Ans (C)**

The Q value of the first reaction implies that

$$^{14}\text{N} + d = ^{15}\text{N} + p + 8.53 \text{ MeV}$$

Where ^{14}N represents the mass of ^{14}N nucleus in energy units.

This can be rearranged to give

$$^{14}\text{N} - ^{15}\text{N} = p - d + 8.53 \text{ MeV}$$

This second reaction similarly implies as

$$^{15}\text{N} - ^{13}\text{C} = p - d + 7.58 \text{ MeV}$$

and the third reaction gives

$$^{13}\text{C} - ^{11}\text{B} = \alpha - d + 5.16 \text{ MeV}$$

adding these three equations, we have

$$^{14}\text{N} - ^{13}\text{C} = p + 2\alpha - 3d + 21.27 \text{ MeV}$$

$$^{11}\text{B} (\alpha, n) ^{14}\text{N} = ^{11}\text{B} - ^{14}\text{N} + \alpha - n$$

$$= 3d - \alpha - p - n - 21.27 \text{ MeV}$$

Now,

$$3d - \alpha - p - n$$

$$= (3 \times 2.014 - 4.0020 - 1.0078 - 1.0087) \text{ a.m.u}$$

$$\therefore Q = (0.0229 \times 931 - 21.27) \text{ MeV} = 0.05 \text{ MeV}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C)**

Conceptual

6. **Ans (A,B,C)**

Conceptual

7. **Ans (A,B,C)**

Initial charge distribution :

$$\left. \begin{array}{c} +\frac{Q}{2} \\ +\frac{Q}{2} \\ -\frac{Q}{2} \end{array} \right| \left. \begin{array}{c} +\frac{Q}{2} \\ -\frac{Q}{2} \end{array} \right| \left. \begin{array}{c} +\frac{Q}{2} \end{array} \right|$$

Maximum current :

$$\frac{Q^2/4}{2\epsilon_0 A/d} = \frac{1}{2} Li^2$$

$$\Rightarrow i = \frac{a}{2} \sqrt{\frac{d}{L\epsilon_0 A}}$$

8. Ans (A,B,D)

$$\frac{P - P/2}{v - 2v} = \frac{P - P_x}{v - v_x}$$

$$\frac{P}{2}(v - v_x) = (P - P_x)v$$

$$2P_x v - P v_x = P v$$

$$2nRvT_x - P v_x^2 = P v v_x$$

$$2P_x^2 v - P nRT_x = P v P_x$$

9. Ans (A,D)

For figure A and B

If $k^2 > R_x$ friction will act in backward direction otherwise in forward direction.

$$\text{For given spool } K^2 = \frac{R^2}{4}$$

$$\text{Figure (a) } R_x = R \cdot \frac{R}{3} = \frac{R^2}{3} \text{ friction forward}$$

$$\text{Figure (b) } R_x = R \cdot \frac{R}{5} = \frac{R^2}{5} \text{ friction backward}$$

In figure (c) as force is below the diametric plane, hence friction backward.

10. Ans (A,C)

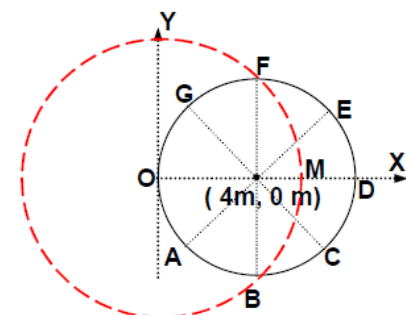
Conceptual

PART-1 : PHYSICS

SECTION-III

2. Ans (8)

Draw a circle assuming OF as radius and O as centre



$$\oint \vec{B} \cdot d\vec{\ell} = \int_M^D \vec{B} \cdot d\vec{\ell} + \int_{DEF} \vec{B} \cdot d\vec{\ell} + \int_F^M \vec{B} \cdot d\vec{\ell} = 0$$

$$\Rightarrow 0 + x - \mu_0 I_0 \frac{45^\circ}{360^\circ} = 0 \Rightarrow x = \frac{\mu_0 I_0}{8}$$

3. Ans (3)

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{6}$$

$$\frac{1}{2}mv^2 - \left(\frac{3}{2}\frac{GM}{R}\right)m = 0 + 0$$

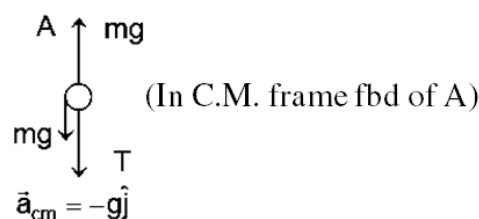
$$\Rightarrow v = \sqrt{\frac{3GM}{R}} = \sqrt{\frac{3}{2}}v_e$$

$$= \sqrt{\frac{3}{2}} \times \sqrt{6}$$

$$\Rightarrow v = 3 \text{ km/s}$$

4. Ans (2)

When Normal contact force approaches zero



In C.M. frame for particle A

$$T = \frac{m\left(\frac{v}{2}\right)^2}{\frac{\ell}{2}} = \frac{mv^2}{2\ell}$$

and for particle B in ground frame

$$\Rightarrow N = mg - T$$

$$= mg - \frac{mv^2}{2\ell}$$

$$\text{particle B will break off when } mg - \frac{mv^2}{2\ell} \leq 0$$

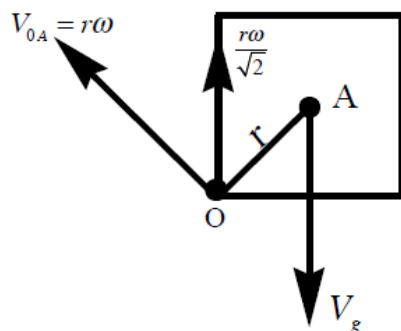
$$\Rightarrow \sqrt{2g\ell} \leq v$$

5. **Ans (5)**

Since $\sum F_x > 0$ $\vec{a}_{cm}$ & $\vec{V}_{cm}$ remains vertical

$$\text{Constraint equation } V_g = \frac{r\omega}{\sqrt{2}} = \frac{I\omega}{2}$$

From L C E



$$\Delta U = \Delta K$$

$$mg \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right) = \frac{1}{2} M V_g^2 + \frac{1}{2} I \omega^2$$

$$mgl \left(\frac{2 - \sqrt{2}}{2\sqrt{2}} \right) = \frac{1}{2} M \left(\frac{I\omega}{2} \right)^2 + \frac{1}{2} \frac{Ml^2}{6} \omega^2$$

$$mgl \left(\frac{2 - \sqrt{2}}{2\sqrt{2}} \right) = \frac{Ml^2 \omega^2}{8} + \frac{ml^2 \omega^2}{12} = \frac{Ml^2 \omega^2}{48}$$

$$\sqrt{\frac{6\sqrt{2}(2 - \sqrt{2})g}{5l}} = \omega$$

6. **Ans (5)**

Conceptual

7. **Ans (7)**

Error in calculated value of speed

$$= \left(\frac{\Delta l}{l} + \frac{\Delta t}{t} \right) \times 100$$

$$= \left(\frac{0.1}{10} + \frac{0.1}{2} \right) \times 100 = 6\%$$

Error in calculated value of distance

$$= \left(\frac{\Delta v}{v} + \frac{\Delta t}{t} \right) \times 100$$

$$= 6\% + \frac{0.1}{10} \times 100 = 7\%$$

Absolute error $\Rightarrow$ distance $\times$ 7%

$$\Rightarrow 50\text{cm} \times 7\% = 3.5 \text{ cm}$$

8. **Ans (1)**

Conceptual

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (C)**

In the given complex: Co^{3+} is d^2sp^3 hybridized. It has more than two N—Co—N bond angles.

2. **Ans (D)**

From concepts of qualitative analysis.

3. **Ans (B)**

Solution is tertiary carbocation followed by ring expansion. pinacol pinacolone type mechanism EAS on benzene followed by bromination and nitration of product obtained.

4. **Ans (B)**

EAS on benzene followed by bromination and nitration of product obtained.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C,D)**

Conceptual

6. **Ans (A,B,C,D)**

In case of B.C.C, nearest distance is $\frac{\sqrt{3}a}{2}$ while next nearest distance is a

7. **Ans (A,B,C)**

In the given double silicates, first one has 2.5 corners shared on average while second one has 3 corners shared.

8. **Ans (A,B,D)**

From the given description, the given metal is Lithium.

9. **Ans (A,B)**

Wurtz reaction; heating produces buta 1,3-diene. Then 1,4 & 1,2 addition

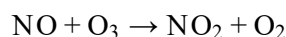
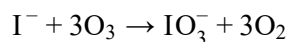
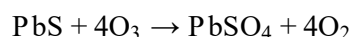
10. **Ans (B,D)**

P is formed by adding of OsO_4 on cis-2-butene Q is formed by addition of mcpBA on trans-2-butene

PART-2 : CHEMISTRY

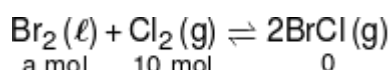
SECTION-III

1. Ans (8)

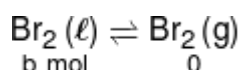


$$x = 4; y = 3; z = 1$$

2. Ans (8)



equ. 0 (10-a)mole 2a mole



equ. 0 b mole

Final moles of $\text{Br}_2(\text{l})$ is taken 0 for minimum

amount of $\text{Br}_2(\text{l})$ needed.

$$b = \frac{0.25 \times 164}{0.082 \times 300} = \frac{5}{3}$$

$$\text{and } (10 + a) = \frac{2 \times 164}{0.082 \times 300} = \frac{40}{3}$$

$$\Rightarrow a = \frac{10}{3}$$

$$\therefore \text{min.mass of Br}_2 = (a + b) \times 160$$

$$= \left(\frac{5}{3} + \frac{10}{3} \right) \times 160 = 800 \text{ gm}$$

3. Ans (3)

$$\text{Given } \Delta T_b = 3.12 \times 10^{-3} \text{ K}$$

$$\text{Solubility of salt} = 2 \times 10^{-3} \text{ mole/kg}$$

$$\Delta T_b = i k_b m$$

$$3.12 \times 10^{-3} = i \times 0.52 \times 2 \times 10^{-3}$$

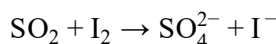
$$i = 3$$

4. Ans (2)



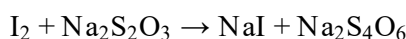
$$t = 0 \quad 0.01 \text{ mol}$$

$$t = 4 \text{ hrs } (0.01 - x) \text{ mol} \quad x \text{ mol } x \text{ mol}$$



$$2x \text{ eq } \frac{25}{100} \text{ eq}$$

$$- \left(\frac{25}{100} - 2x \right) \text{ eq}$$



$$\frac{25}{100} - 2X = \frac{10}{1000} \rightarrow X = 0.0075 \text{ mol}$$

In case of 1st order kinetics

$$K = \frac{0.693}{t_{1/2}} = \frac{2.303}{t} \log \frac{a}{a-X}$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{4} \log \frac{0.01}{0.01 - 0.0075}$$

$$t_{1/2} = 2 \text{ hrs}$$

5. Ans (3)

$$\text{Given } R = 200 \Omega$$

$$\text{Cell constant} = 4 \text{ cm}^{-1}$$

$$K = \frac{1}{R} \times \frac{1}{a} = \frac{1}{200} \times 4 = \frac{1}{50} \Omega^{-1} \text{ cm}^{-1}$$

$$\lambda_m = K \times \frac{1000}{M} = \frac{1}{50} \times \frac{1000}{0.01}$$

$$= \frac{10^5}{50} = 2 \times 10^3$$

$$\alpha = \frac{\lambda_m}{\lambda_m^0} = \frac{2 \times 10^3}{2 \times 10^4} = 0.01$$

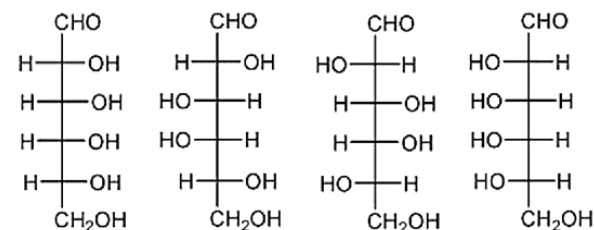
$$[\text{H}^+] = C\alpha = 0.01 \times 0.1 = 10^{-3} \text{ m}$$

$$\text{p}^{\text{H}} = 3$$

6. Ans (5)

Gold sol, clay, Acidic dye, starch and As_2S_3 sols are negatively charged

7. Ans (4)



8. Ans (5)

acylation at $-\text{NH}_2$ group of P-amino phenol

PART-3 : MATHEMATICS

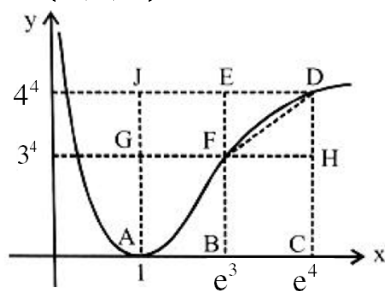
SECTION-I (ii)

6. **Ans (A,B,C)**

Let $2^x = a$ and $3^x = b$

$$\begin{aligned} \Rightarrow 2^x + 3^x - 4^x + 6^x - 9^x &= a + b - a^2 + ab - b^2 \\ &= -(a^2 - ab + b^2 - a - b + 1) + 1 \\ &= -\frac{1}{2}((a-b)^2 + (b-1)^2 + (a-1)^2) + 1 \leq 1 \end{aligned}$$

7. **Ans (A,C,D)**



A is area bounded by curve between $x = 1$, $x = e^4$

$x = e^3$ is point of inflection

Now, $A < \text{Ar.}(\square ABGF) + \text{Ar.}(\square BCDE)$

$$\Rightarrow A(e^3 - 1)3^4 + (e^4 - e^3)4^4$$

Again, $A > \text{Ar.}(\text{trapezium BFDC})$

$$\Rightarrow A > \frac{1}{2}(3^4 + 4^4)(e^4 - e^3)$$

$$\Rightarrow A > \frac{337}{2}(e^3(e-1))$$

8. **Ans (C,D)**

Since, $x^4 + y^4 + z^4 + 1 = 4xyz$

$$\Rightarrow xyz \geq 0$$

So, two cases arise : (i) $x, y, z \geq 0$

(ii) out of $x, y, z \geq 0$

$$\therefore \text{A.M.} \geq \text{G.M.} \Rightarrow \frac{x^4 + y^4 + z^4 + 1}{4} \geq xyz$$

So, equality must hold $\Rightarrow x = y = z = 1$

Hence, vertices are

$(1,1,1), (1,-1,-1), (-1,1,-1)$ & $(-1,-1,1)$

$$\text{So, } v = \frac{1}{6} \begin{vmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{vmatrix} = \frac{8}{3}$$

9. **Ans (A,B,D)**

$$P(\bar{A}|\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cap B)}{1 - P(B)} = \frac{4}{5}$$

$$P(A \cap \bar{B} \cap C) + P(A \cap \bar{B} \cap \bar{C}) = P(A \cap \bar{B}) = 1/7$$

$$P(A \cap B) = P(A) - P(A \cap \bar{B}) = 1/7$$

10. **Ans (B,C)**

$$\begin{aligned} &\int_0^{1/a^2} \frac{\cos^{-1} ax}{\sec^{-1}(ax+2)} dx \\ &\lim_{a \rightarrow \infty} \frac{\int_0^{1/a^2} \frac{\cos^{-1} ax}{\sec^{-1}(ax+2)} dx}{\frac{1}{a^2}}, \text{ let } ax = t \\ &= \lim_{a \rightarrow \infty} \frac{\int_0^{1/a} \frac{\cos^{-1} t}{\sec^{-1}(t+2)} dt}{\frac{1}{a}} \\ &= \lim_{a \rightarrow \infty} \frac{\cos^{-1} \frac{1}{a}}{\sec^{-1} \left(\frac{1}{a} + 2 \right)} = \frac{\pi/2}{\pi/3} = \frac{3}{2} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (7)**

$$A^2 = I, \lambda = 4$$

$$\therefore B^2 = 81I$$

$$\therefore \text{Tr} \left(\frac{A^2 + B^2}{82} \right) + \lambda = 7$$

3. **Ans (5)**

Let point on hyperbola $(3 \sec t, 2 \tan t)$ equation of chord of contact

$$x(3 \sec t) + y(2 \tan t) = 9$$

$$\text{Compare with } xh + yk - h^2 - k^2 = 0$$

$$\frac{3 \sec t}{h} = \frac{2 \tan t}{k} = \frac{9}{h^2 + k^2}$$

$$\sec t = \frac{3h}{h^2 + k^2}, \tan t = \frac{9}{2} \frac{k}{h^2 + k^2}$$

$$\sec^2 t - \tan^2 t = \frac{9h^2}{(h^2 + k^2)^2} - \frac{81}{4} \frac{k^2}{(h^2 + k^2)^2}$$

$$4(h^2 + k^2) = 36h^2 - 81k^2$$

$$\alpha = 4$$

$$\beta = 36$$

$$\Rightarrow 81 - 72 - 4$$

$$\gamma = 81 = 5$$

4. Ans (5)

$$\lim_{x \rightarrow \infty} x^4 \left(e^{x^2 \ln \left(\frac{x^2+1}{x^2-1} \right)} - e^2 \right)$$

$$e^2 \lim_{x \rightarrow \infty} x^4 \left(e^{x^2 \ln \left(\frac{x^2+1}{x^2-1} \right) - 2} - 1 \right)$$

$$e^2 \lim_{x \rightarrow \infty} x^4 \left(x^2 \ln \left(\frac{x^2+1}{x^2-1} \right) - 2 \right)$$

$$\text{Put } x = \frac{1}{y}$$

$$e^2 \lim_{y \rightarrow 0} \left(\frac{\ln(1+y^2) - \ln(1-y^2) - 2y^2}{y^6} \right)$$

$$= \frac{2}{3} e^2$$

(use expansion)

$$\therefore |a^2 - b^2| = 5$$

5. Ans (1)

$$\begin{aligned} & \frac{1}{\cot 9^\circ - 3 \tan 9^\circ} + \frac{3}{\cot 27^\circ - 3 \tan 27^\circ} + \\ & \frac{9}{\cot 81^\circ - 3 \tan 81^\circ} + \frac{27}{\cot 243^\circ - 3 \tan 243^\circ} \\ &= \frac{\tan 9^\circ}{1 - 3 \tan^2 9^\circ} + \frac{3 \tan 27^\circ}{1 - 3 \tan^2 27^\circ} + \\ & \frac{9 \tan 81^\circ}{1 - 3 \tan^2 81^\circ} + \frac{27 \tan 243^\circ}{1 - 3 \tan^2 243^\circ} \\ &= \left(\frac{3 \tan 27^\circ - \tan 9^\circ}{8} \right) + 3 \left(\frac{3 \tan 81^\circ - \tan 27^\circ}{8} \right) \\ &+ 9 \left(\frac{3 \tan 243^\circ - \tan 81^\circ}{8} \right) + 27 \left(\frac{3 \tan 729^\circ - \tan 243^\circ}{8} \right) \\ &= \frac{81 \tan 729^\circ - \tan 9^\circ}{8} = 10 \tan 9^\circ \end{aligned}$$

$$\therefore A = 10, B = 9$$

$$\therefore |A - B| = 1$$

6. Ans (1)

$$f - f' = f' - f''$$

$$c+x = \ln(f - f')$$

$$f - f' = Ae^x$$

$$\Rightarrow \frac{dy}{dx} - y = -Ae^x$$

$$y \cdot e^{-x} = \int -Ae^x \cdot e^{-x} dx$$

$$ye^{-x} = -Ax + B$$

$$y(0)=0, y(1)=e$$

$$\Rightarrow y = xe^x \Rightarrow \int_0^1 xe^x dx = 1$$

7. Ans (1)

$$\tan x > 0 \text{ \& } \tan x \neq 1$$

$$\left(\log_{\sqrt{3}} \tan x \right) \left(\sqrt{\frac{\log_{\sqrt{3}} (\sqrt{3})^3}{\log_{\sqrt{3}} \sqrt{3}}} + \frac{\log_{\sqrt{3}} 3}{\log_{\sqrt{3}} \tan x} \right) = -1$$

$$\text{Let } \log_{\sqrt{3}} \tan x = t < 0$$

$$\Rightarrow t \sqrt{3 + \frac{2}{t}} = -1$$

$$\Rightarrow 3t^2 + 2t - 1 = 0$$

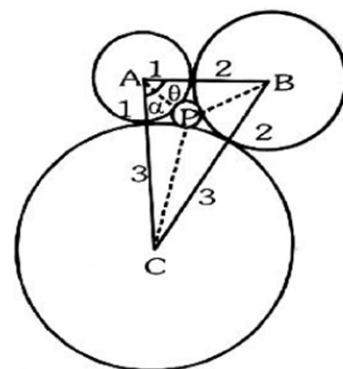
$$\Rightarrow t = -1 \text{ \& } t = \frac{1}{2} \text{ (not possible)}$$

$$\therefore \log_{\sqrt{3}} \tan x = -1 \Rightarrow x = \frac{\pi}{6} \text{ only in } [0, \pi]$$

8. Ans (6)

ΔABC is right angle

Applying cosine rule in ΔPAB



$$\cos \theta = \frac{3^2 + (1+r)^2 - (2+r)^2}{2 \cdot 3(1+r)} = \frac{3-r}{3(1+r)}$$

Again applying cosine rule in ΔPAC

$$\cos \alpha = \frac{(1+r)^2 + 4^2 - (3+r)^2}{2 \cdot 4(1+r)} = \frac{2-r}{2(1+r)}$$

$$\therefore \alpha + \theta = 90$$

$$\alpha = 90 - \theta \Rightarrow \cos \alpha = \sin \theta$$

$$\left(\frac{3-r}{3(r+1)} \right)^2 + \left(\frac{2-r}{2(r+1)} \right)^2 = 1$$